# CSCI3230 / ESTR3108 2026–27 First Term Assignment 1

*I declare that the assignment here submitted is original except for source material explicitly acknowledged, and that the same or closely related material has not been previously submitted for another course. I also acknowledge that I am aware of University policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the following websites.*

University Guideline on Academic Honesty:
`http://www.cuhk.edu.hk/policy/academichonesty/`
Faculty of Engineering Guidelines to Academic Honesty:
`http://www.erg.cuhk.edu.hk/erg-intra/upload/documents/ENGG_Discipline.pdf`

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*Note.* All numerical results were computed in Python with NumPy and independently cross-checked: the normal-equation solutions against `numpy.linalg.lstsq` and `numpy.polyfit`, and the logistic-regression gradient against a central finite-difference approximation.

## Question 1: Linear models and model complexity

### (a) $\hat{f}_{\boldsymbol{\Theta}}(x) = \theta_0 + \theta_1 x$

With the design matrix and parameter vector

$$\mathbf{X} = \begin{pmatrix} 1 & x^{(1)} \\ \vdots & \vdots \\ 1 & x^{(10)} \end{pmatrix} \in \mathbb{R}^{10\times 2}, \qquad \boldsymbol{\Theta} = \begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix},$$

the RSS objective is $J(\boldsymbol{\Theta}) = \|\mathbf{X}\boldsymbol{\Theta} - \mathbf{y}\|_2^2$, a convex quadratic in $\boldsymbol{\Theta}$. Setting its gradient to zero gives the normal equations

$$\mathbf{X}^\top \mathbf{X}\,\boldsymbol{\Theta} = \mathbf{X}^\top \mathbf{y} \qquad \Longrightarrow \qquad \hat{\boldsymbol{\Theta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{y}.$$

From the training data: $\sum_i x_i = 56.1$, $\sum_i x_i^2 = 353.09$, $\sum_i y_i = 8.5$, $\sum_i x_i y_i = 52.88$, so

$$\begin{pmatrix} 10 & 56.1 \\ 56.1 & 353.09 \end{pmatrix} \begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix} = \begin{pmatrix} 8.5 \\ 52.88 \end{pmatrix}.$$

Solving ($10 \cdot 353.09 - 56.1^2 = 383.69$):

$$\theta_1 = \frac{10 \cdot 52.88 - 56.1 \cdot 8.5}{383.69} = \frac{51.950}{383.69} = 0.135396, \qquad \theta_0 = \bar{y} - \theta_1 \bar{x} = 0.85 - 0.135396 \cdot 5.61 = 0.090430.$$

$$\boxed{\hat{f}(x) = 0.0904 + 0.1354\,x} \qquad \text{(training RSS} = 4.7816\text{)}$$

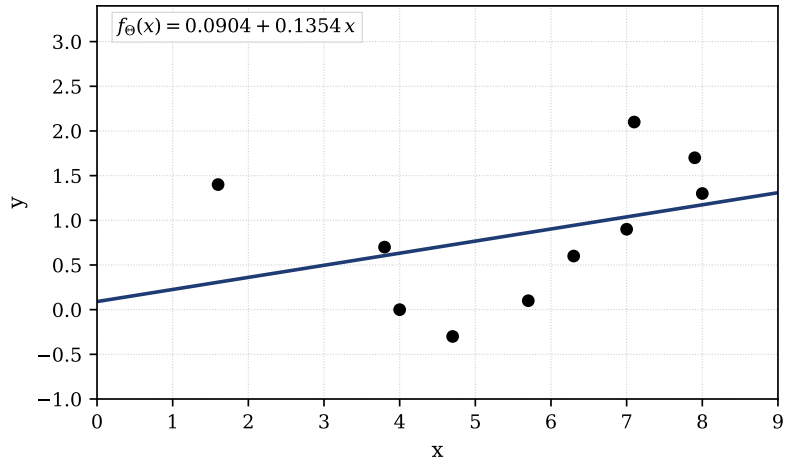



Figure 1: Training data and the fitted line $\hat{f}(x) = 0.0904 + 0.1354x$.

**(b)** $\hat{f}_{\boldsymbol{\Theta}}(x) = \theta_0 + \theta_1 x + \theta_2 x^2$

Now each row of $\mathbf{X}$ is $(1,\ x^{(i)},\ (x^{(i)})^2)$ and $\boldsymbol{\Theta} = (\theta_0, \theta_1, \theta_2)^\top$. The same normal equations $\mathbf{X}^\top\mathbf{X}\boldsymbol{\Theta} = \mathbf{X}^\top\mathbf{y}$ read

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 10 & 56.1 & 353.09 \\ 56.1 & 353.09 & 2367.981 \\ 353.09 & 2367.981 & 16523.1077 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{y} = \begin{pmatrix} 8.5 \\ 52.88 \\ 373.386 \end{pmatrix}.$$

Solving this $3 \times 3$ system,

$$\hat{\boldsymbol{\Theta}} = (3.058928,\ -1.271061,\ 0.139390)^\top \qquad \Longrightarrow \qquad \boxed{\hat{f}(x) = 3.0589 - 1.2711\,x + 0.1394\,x^2}$$

with training RSS $= 1.8765$.

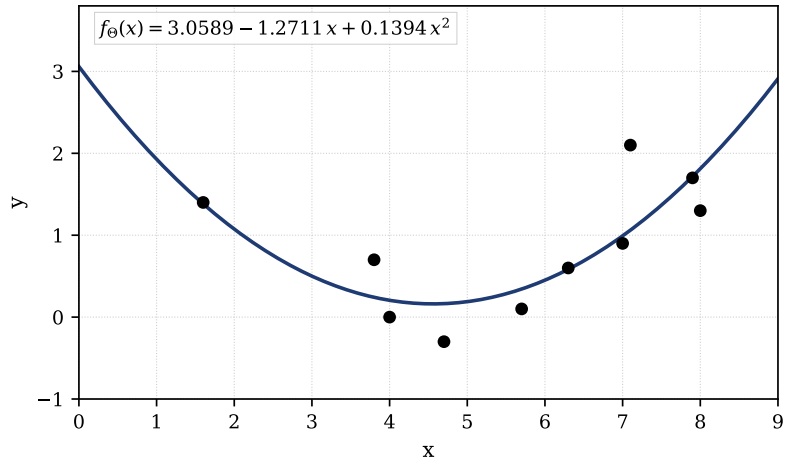



Figure 2: Training data and the fitted parabola $\hat{f}(x) = 3.0589 - 1.2711x + 0.1394x^2$.

**(c)** $\hat{f}_{\boldsymbol{\Theta}}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

Each row of $\mathbf{X}$ is $(1,\ x^{(i)},\ (x^{(i)})^2,\ (x^{(i)})^3,\ (x^{(i)})^4)$ and $\boldsymbol{\Theta} = (\theta_0, \ldots, \theta_4)^\top$. Solving the corresponding $5 \times 5$ normal equations (solved numerically, as permitted for calculations):

$$\hat{\boldsymbol{\Theta}} = (-12.176904,\ 16.445766,\ -6.377749,\ 0.954652,\ -0.048479)^\top$$

$$\boxed{\hat{f}(x) = -12.1769 + 16.4458\,x - 6.3777\,x^2 + 0.9547\,x^3 - 0.0485\,x^4} \qquad \text{(training RSS} = 0.7712\text{)}$$

The quartic no longer follows a smooth bowl: it dips to about $-0.34$ around $x \approx 5$ and (as seen in Figure 3) swings down to $-12.2$ at $x = 0$, far outside the range of the data.

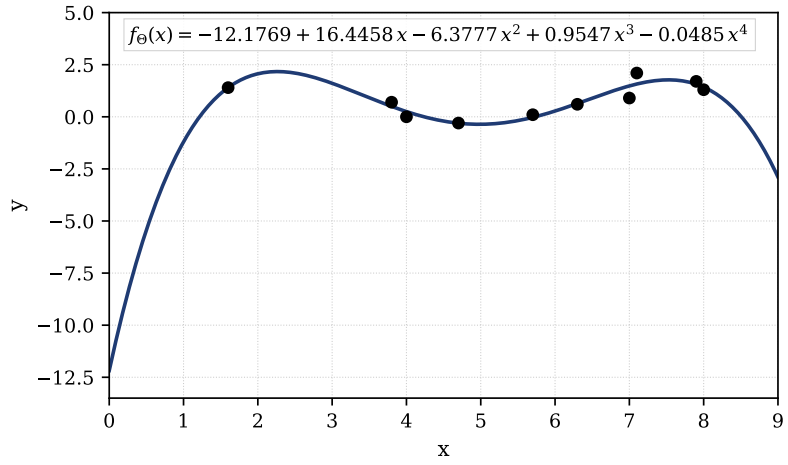



Figure 3: Training data and the fitted quartic $\hat{f}(x) = -12.1769 + 16.4458x - 6.3777x^2 + 0.9547x^3 - 0.0485x^4$.

## (d) Underfitting, overfitting, and test error

Table 1: Train / test fit quality of the three models (error sums over the 10 points of each set).

| Model | Train RSS | Train MSE | Test RSS | Test MSE |
|---|---|---|---|---|
| Degree 1 (line) | 4.7816 | 0.4782 | 4.1244 | 0.4124 |
| Degree 2 (parabola) | 1.8765 | 0.1877 | 2.4768 | 0.2477 |
| Degree 4 (quartic) | 0.7712 | 0.0771 | 5.1835 | 0.5184 |

**Which model does what.**

- **Degree 1 — underfitting.** The training points rise, bend over and rise again (a convex "bowl"), but any straight line can only cross that pattern: the residuals in Figure 1(a) are systematically negative in the middle and positive at both ends. The line cannot represent the curvature, so *both* the training error (RSS = 4.78) and the test error (RSS = 4.12) stay large — the hallmark of *high bias*. (Its test RSS happens to be slightly below its training RSS in this particular sample; that is coincidence of where the test points fall, not evidence of a good fit — both errors are about twice as large as the parabola's.)

Table 2: Per-point test-set predictions and squared errors.

| $x$ | $y$ | $f_1(x)$ | $e_1^2$ | $f_2(x)$ | $e_2^2$ | $f_4(x)$ | $e_4^2$ |
|---|---|---|---|---|---|---|---|
| 7.2 | 0.6 | 1.0653 | 0.2165 | 1.1333 | 0.2844 | 1.6514 | 1.1055 |
| 1.1 | 1.0 | 0.2394 | 0.5786 | 1.8294 | 0.6879 | −0.6040 | 2.5727 |
| 3.6 | 0.3 | 0.5779 | 0.0772 | 0.2896 | 0.0001 | 0.7699 | 0.2208 |
| 4.8 | −0.4 | 0.7403 | 1.3004 | 0.1694 | 0.3242 | −0.3381 | 0.0038 |
| 4.3 | 0.0 | 0.6726 | 0.4524 | 0.1707 | 0.0291 | −0.0570 | 0.0033 |
| 6.6 | 0.2 | 0.9840 | 0.6147 | 0.7418 | 0.2935 | 1.0222 | 0.6760 |
| 7.8 | 1.8 | 1.1465 | 0.4270 | 1.6251 | 0.0306 | 1.6661 | 0.0179 |
| 5.6 | 0.6 | 0.8486 | 0.0618 | 0.3123 | 0.0828 | −0.1109 | 0.5054 |
| 7.3 | 1.7 | 1.0788 | 0.3859 | 1.2083 | 0.2418 | 1.7122 | 0.0001 |
| 8.2 | 1.3 | 1.2007 | 0.0099 | 2.0088 | 0.5024 | 1.0209 | 0.0779 |
| Test RSS | | | 4.1244 | | 2.4768 | | 5.1835 |

- **Degree 4 — overfitting.** The quartic has enough flexibility to chase every training point: the smallest training RSS (0.77) of the three. But the extra wiggles are fitted to the noise, not to the trend — at $x = 1.1$ it predicts $-0.60$ while the true label is 1.0 (squared error 2.57, the largest single term in the table), and it blows up to $-12.2$ at $x = 0$. On unseen data its error is the *largest* (5.18, versus 4.78 for the line): classic *overfitting* (high variance).
- **Degree 2 — the good model.** It captures the convex trend with just three parameters, and it achieves the *smallest test error* (RSS = 2.48, MSE = 0.248) while its training error (1.88) stays moderate. Its test error is only slightly above its training error, the normal, healthy pattern of a model of the right complexity.

In short: more parameters monotonically lower the training error ($4.78 \to 1.88 \to 0.77$), but the test error first falls and then rises ($4.12 \to 2.48 \to 5.18$) — the classic bias–variance trade-off. The degree-2 model is the one that generalizes best, so it is "relatively the good one".

## Question 2: Analytic solution of ridge regression

Notation: $m$ samples $x^{(i)} \in \mathbb{R}^n$, labels $Y^{(i)}$, parameter vector $\mathbf{\Theta} \in \mathbb{R}^n$, intercept $\theta_0$. Let $\bar{\mathbf{x}} = \frac{1}{m}\sum_i x^{(i)}$ and $\bar{Y} = \frac{1}{m}\sum_i Y^{(i)}$ denote the means, so $\sum_i (x^{(i)} - \bar{\mathbf{x}}) = 0$ and $\sum_i (Y^{(i)} - \bar{Y}) = 0$.

### (a) Optimality of $\theta_0 = \bar{Y} - \bar{\mathbf{x}}^\top \mathbf{\Theta}$

Write, for each sample, $g_i = (x^{(i)} - \bar{\mathbf{x}})^\top \mathbf{\Theta} + \theta_0 + \bar{\mathbf{x}}^\top \mathbf{\Theta} - Y^{(i)}$, so that $J(\mathbf{\Theta}, \theta_0) = \sum_{i=1}^m g_i^2 + \lambda \|\mathbf{\Theta}\|_2^2$. Differentiating with respect to the scalar $\theta_0$ (the regularizer does not involve $\theta_0$):

$$\frac{\partial J}{\partial \theta_0} = 2\sum_{i=1}^m g_i = 2\Big[ \underbrace{\sum_i (x^{(i)} - \bar{\mathbf{x}})^\top}_{=0} \mathbf{\Theta} \;+\; m\theta_0 \;+\; m\,\bar{\mathbf{x}}^\top \mathbf{\Theta} - \sum_i Y^{(i)} \Big] = 2m\big(\theta_0 + \bar{\mathbf{x}}^\top \mathbf{\Theta} - \bar{Y}\big),$$

where we used $\sum_i x^{(i)} = m\bar{\mathbf{x}}$, hence $\sum_i (x^{(i)} - \bar{\mathbf{x}}) = 0$, and $\sum_i Y^{(i)} = m\bar{Y}$. Setting this to zero gives

$$\theta_0 = \bar{Y} - \bar{\mathbf{x}}^\top \mathbf{\Theta}.$$

Furthermore $\partial^2 J / \partial \theta_0^2 = 2m > 0$, so for any fixed $\mathbf{\Theta}$ the function is strictly convex in $\theta_0$ and this stationary point is the *unique global minimum*. □

### (b) Reduction to the centered problem

Plug $\theta_0 = \bar{Y} - \bar{\mathbf{x}}^\top \boldsymbol{\Theta}$ into each term:

$$g_i = (x^{(i)} - \bar{\mathbf{x}})^\top \boldsymbol{\Theta} + \big(\bar{Y} - \bar{\mathbf{x}}^\top \boldsymbol{\Theta}\big) + \bar{\mathbf{x}}^\top \boldsymbol{\Theta} - Y^{(i)} = (x^{(i)} - \bar{\mathbf{x}})^\top \boldsymbol{\Theta} - \big(Y^{(i)} - \bar{Y}\big) = \big(x_c^{(i)}\big)^\top \boldsymbol{\Theta} - Y_c^{(i)},$$

with the centered input $x_c^{(i)} = x^{(i)} - \bar{\mathbf{x}}$ and centered label $Y_c^{(i)} = Y^{(i)} - \bar{Y}$. Therefore

$$J(\boldsymbol{\Theta}, \theta_0) = \sum_{i=1}^{m} \Big[\big(x_c^{(i)}\big)^\top \boldsymbol{\Theta} - Y_c^{(i)}\Big]^2 + \lambda \|\boldsymbol{\Theta}\|_2^2 = J_c(\boldsymbol{\Theta}). \square$$

### (c) Analytic solution

Let $\mathbf{X}_c \in \mathbb{R}^{m \times n}$ be the matrix whose $i$-th row is $(x_c^{(i)})^\top$ and $\mathbf{Y}_c = (Y_c^{(1)}, \ldots, Y_c^{(m)})^\top$, so $J_c(\boldsymbol{\Theta}) = \|\mathbf{X}_c\boldsymbol{\Theta} - \mathbf{Y}_c\|_2^2 + \lambda\|\boldsymbol{\Theta}\|_2^2 = \boldsymbol{\Theta}^\top \mathbf{X}_c^\top \mathbf{X}_c \boldsymbol{\Theta} - 2\boldsymbol{\Theta}^\top \mathbf{X}_c^\top \mathbf{Y}_c + \text{const}$. Differentiating with respect to the vector $\boldsymbol{\Theta}$ and setting the gradient to zero:

$$\frac{\partial J_c}{\partial \boldsymbol{\Theta}} = 2\mathbf{X}_c^\top \big(\mathbf{X}_c\boldsymbol{\Theta} - \mathbf{Y}_c\big) + 2\lambda\boldsymbol{\Theta} = 0 \qquad \Longrightarrow \qquad \big(\mathbf{X}_c^\top \mathbf{X}_c + \lambda I_n\big)\boldsymbol{\Theta} = \mathbf{X}_c^\top \mathbf{Y}_c.$$

Since $\lambda > 0$, for every $\boldsymbol{\Theta} \neq 0$, $\boldsymbol{\Theta}^\top(\mathbf{X}_c^\top \mathbf{X}_c + \lambda I_n)\boldsymbol{\Theta} = \|\mathbf{X}_c\boldsymbol{\Theta}\|_2^2 + \lambda\|\boldsymbol{\Theta}\|_2^2 > 0$, so the matrix is positive definite and hence invertible; $J_c$ is strictly convex and the solution is unique:

$$\boxed{\hat{\boldsymbol{\Theta}} = \big(\mathbf{X}_c^\top \mathbf{X}_c + \lambda I_n\big)^{-1} \mathbf{X}_c^\top \mathbf{Y}_c}$$

and the intercept follows from part (a): $\hat{\theta}_0 = \bar{Y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\Theta}}$. $\square$

## Question 3: Logistic regression by gradient descent

### (a) Model and cross-entropy error

With $f_{\boldsymbol{\Theta}}(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$, the logistic model is

$$P(\hat{y} = 1 \mid x_1, x_2) = \sigma\big(f_{\boldsymbol{\Theta}}(x_1, x_2)\big) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}}, \qquad P(\hat{y} = 0 \mid x_1, x_2) = 1 - \sigma\big(f_{\boldsymbol{\Theta}}(x_1, x_2)\big),$$

and the cross-entropy error function is

$$E(\boldsymbol{\Theta}) = -\sum_{i=1}^{6} \Big[\, y^{(i)} \ln \sigma\big(f_{\boldsymbol{\Theta}}(x^{(i)})\big) + \big(1 - y^{(i)}\big) \ln\Big(1 - \sigma\big(f_{\boldsymbol{\Theta}}(x^{(i)})\big)\Big)\Big].$$

### (b) One gradient-descent step

Since $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, the gradient of $E$ is the familiar

$$\frac{\partial E}{\partial \theta_j} = \sum_{i=1}^{6} \Big(\sigma\big(f_{\boldsymbol{\Theta}}(x^{(i)})\big) - y^{(i)}\Big)\, x_j^{(i)}, \qquad j = 0, 1, 2 \quad (x_0^{(i)} \equiv 1).$$

With the initial weights $\boldsymbol{\Theta}^{(0)} = (-1,\, 1.5,\, 0.5)^\top$, $z^{(i)} = \theta_0 + \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)}$:

Table 3: Training set: linear outputs $z^{(i)}$, probabilities $\sigma(z^{(i)})$ and residuals.

| $i$ | $x_1$ | $x_2$ | $y$ | $z$ | $\sigma(z)$ | $\sigma(z) - y$ |
|---|---|---|---|---|---|---|
| 1 | 0.15 | 0.35 | 0 | $-0.6000$ | 0.3543 | 0.3543 |
| 2 | 0.30 | 0.45 | 0 | $-0.3250$ | 0.4195 | 0.4195 |
| 3 | 0.55 | 0.25 | 0 | $-0.0500$ | 0.4875 | 0.4875 |
| 4 | 0.30 | 0.80 | 1 | $-0.1500$ | 0.4626 | $-0.5374$ |
| 5 | 0.55 | 0.60 | 1 | 0.1250 | 0.5312 | $-0.4688$ |
| 6 | 0.70 | 0.50 | 1 | 0.3000 | 0.5744 | $-0.4256$ |

Summing the residuals weighted by each input:

$$\frac{\partial E}{\partial \theta_0} = \sum_i \big(\sigma(z^{(i)}) - y^{(i)}\big) = -0.1705,$$
$$\frac{\partial E}{\partial \theta_1} = \sum_i \big(\sigma(z^{(i)}) - y^{(i)}\big) x_1^{(i)} = -0.2698,$$
$$\frac{\partial E}{\partial \theta_2} = \sum_i \big(\sigma(z^{(i)}) - y^{(i)}\big) x_2^{(i)} = -0.4893.$$

The gradient-descent update with learning rate $\eta = 0.1$ is $\boldsymbol{\Theta}^{(1)} = \boldsymbol{\Theta}^{(0)} - 0.1\,\nabla E(\boldsymbol{\Theta}^{(0)})$:

$$\boldsymbol{\Theta}^{(1)} = (-1,\, 1.5,\, 0.5)^\top - 0.1\,(-0.1705,\, -0.2698,\, -0.4893)^\top = (-0.98295,\, 1.52698,\, 0.54893)^\top.$$

$$\boxed{\hat{f}_{\boldsymbol{\Theta}^{(1)}}(x_1, x_2) = -0.98295 + 1.52698\, x_1 + 0.54893\, x_2}$$

As a sanity check, the loss decreases from $E(\boldsymbol{\Theta}^{(0)}) = 3.60765$ to $E(\boldsymbol{\Theta}^{(1)}) = 3.57563$.

### (c) Test-set predictions, accuracy, precision, recall

Using the updated model ($z$ and $\sigma(z)$ computed on the test set; $\hat{y} = 1$ iff $\sigma \geq 0.5$):

Table 4: Test set: predictions of the updated model.

| $i$ | $x_1$ | $x_2$ | $y$ | $z$ | $\sigma(z)$ | $\hat{y}$ |
|---|---|---|---|---|---|---|
| 1 | 0.28 | 0.82 | 1 | $-0.1053$ | 0.4737 | 0 |
| 2 | 0.45 | 0.75 | 1 | 0.1159 | 0.5289 | 1 |
| 3 | 0.58 | 0.62 | 1 | 0.2430 | 0.5605 | 1 |
| 4 | 0.72 | 0.48 | 1 | 0.3800 | 0.5939 | 1 |
| 5 | 0.22 | 0.40 | 0 | $-0.4274$ | 0.3947 | 0 |
| 6 | 0.50 | 0.26 | 0 | $-0.0767$ | 0.4808 | 0 |

Confusion counts: TP = 3 (samples 2–4), FP = 0, FN = 1 (sample 1), TN = 2 (samples 5–6). Hence

$$\text{accuracy} = \frac{TP + TN}{6} = \frac{5}{6} \approx 0.8333, \qquad \text{precision} = \frac{TP}{TP + FP} = \frac{3}{3} = 1.000, \qquad \text{recall} = \frac{TP}{TP + FN} = \frac{3}{4} = 0.75$$

The only error is sample 1, $(x_1, x_2) = (0.28, 0.82)$ with label 1: it was already the hardest training-like point (large $x_2$ but very small $x_1$), and after one step $\sigma = 0.474 < 0.5$, so the classifier still

(barely) predicts 0. Precision is perfect because no negative sample is mislabelled positive; recall is 3/4 because of this one missed positive.

## Question 4: Softmax derivatives

For a fixed sample $\mathbf{x}$ (with linear scores $s_k = \mathbf{x}^\top \boldsymbol{\Theta}_k$) write

$$P_i := P(\hat{y}_i = 1 \mid \mathbf{x}) = \frac{e^{s_i}}{\sum_{k=1}^{K} e^{s_k}}, \qquad i = 1, \dots, K.$$

### (a) Derivative with respect to $\mathbf{x}^\top \boldsymbol{\Theta}_j$

The denominator $S := \sum_{k=1}^{K} e^{s_k}$ depends on $s_j$ for every $j$.

*Case $j = i$:* differentiating $P_i = e^{s_i}/S$ with the quotient rule,

$$\frac{\partial P_i}{\partial s_i} = \frac{e^{s_i} S - e^{s_i}\, e^{s_i}}{S^2} = P_i - P_i^2 = P_i\,(1 - P_i) = P(\hat{y}_i = 1 \mid \mathbf{x}) \Big(1 - P(\hat{y}_i = 1 \mid \mathbf{x})\Big).$$

*Case $j \neq i$:* in the numerator $e^{s_i}$ the variable $s_j$ does not appear, and $\partial S/\partial s_j = e^{s_j}$, so

$$\frac{\partial P_i}{\partial s_j} = \frac{0 - e^{s_i} e^{s_j}}{S^2} = -\,P_i P_j = -\,P(\hat{y}_i = 1 \mid \mathbf{x})\, P(\hat{y}_j = 1 \mid \mathbf{x}).$$

Combining the two cases,

$$\frac{\partial P(\hat{y}_i = 1 \mid \mathbf{x})}{\partial \mathbf{x}^\top \boldsymbol{\Theta}_j} = \begin{cases} P(\hat{y}_i = 1 \mid \mathbf{x})\big(1 - P(\hat{y}_i = 1 \mid \mathbf{x})\big), & j = i, \\ -\,P(\hat{y}_i = 1 \mid \mathbf{x})\, P(\hat{y}_j = 1 \mid \mathbf{x}), & j \neq i. \end{cases} \qquad \square$$

### (b) Derivative with respect to $\boldsymbol{\Theta}_j$

By the chain rule (the hint),

$$\frac{\partial P_i}{\partial \boldsymbol{\Theta}_j} = \underbrace{\frac{\partial P_i}{\partial \big(\mathbf{x}^\top \boldsymbol{\Theta}_j\big)}}_{\text{part (a)}} \cdot \underbrace{\frac{\partial \big(\mathbf{x}^\top \boldsymbol{\Theta}_j\big)}{\partial \boldsymbol{\Theta}_j}}_{=\,\mathbf{x}},$$

since $\mathbf{x}$ is a fixed sample vector and $\mathbf{x}^\top \boldsymbol{\Theta}_j$ is linear in $\boldsymbol{\Theta}_j$. Therefore

$$\frac{\partial P(\hat{y}_i = 1 \mid \mathbf{x})}{\partial \boldsymbol{\Theta}_j} = \begin{cases} \mathbf{x}\, P(\hat{y}_i = 1 \mid \mathbf{x})\big(1 - P(\hat{y}_i = 1 \mid \mathbf{x})\big), & j = i, \\ -\,\mathbf{x}\, P(\hat{y}_i = 1 \mid \mathbf{x})\, P(\hat{y}_j = 1 \mid \mathbf{x}), & j \neq i. \end{cases} \qquad \square$$

### (c) Gradient of the negative log-likelihood

With the one-hot labels $y_i$ ($\sum_i y_i = 1$) and $E(\boldsymbol{\Theta}) = -\sum_i y_i \ln P_i$:

$$\frac{\partial E}{\partial \boldsymbol{\Theta}_j} = -\sum_{i=1}^{K} y_i \,\frac{1}{P_i}\, \frac{\partial P_i}{\partial \boldsymbol{\Theta}_j} = -\sum_{i=1}^{K} y_i \,\frac{1}{P_i} \cdot \mathbf{x}\, \frac{\partial P_i}{\partial s_j},$$

where we inserted the chain rule of part (b), $\partial P_i/\partial \boldsymbol{\Theta}_j = \mathbf{x}\,\partial P_i/\partial s_j$. Using part (a) in the compact form $\partial P_i/\partial s_j = P_i\,(\delta_{ij} - P_j)$, with $\delta_{ij}$ the Kronecker delta,

$$\frac{\partial E}{\partial \boldsymbol{\Theta}_j} = -\sum_{i=1}^{K} y_i\,\frac{1}{P_i}\,P_i\,(\delta_{ij} - P_j)\,\mathbf{x} = -\,\mathbf{x}\sum_{i=1}^{K} y_i\,(\delta_{ij} - P_j) = -\,\mathbf{x}\Big(\underbrace{\sum_i y_i\delta_{ij}}_{=\,y_j} - P_j\underbrace{\sum_i y_i}_{=\,1}\Big),$$

and therefore

$$\boxed{\frac{\partial E(\boldsymbol{\Theta})}{\partial \boldsymbol{\Theta}_j} = \mathbf{x}\,P(\hat{y}_j = 1 \mid \mathbf{x}) - \mathbf{x}\,y_j}$$

as required. This is the softmax counterpart of the logistic-regression gradient $(\sigma - y)$ in Question 3. □